



Network Management

Understanding IPv4 Networking



- In IPv4, each node needs its own IP address, written in dotted decimal notation
 - E.g. 192.168.4.22/24 → used to calculate network address
- Each IP address must be indicated with the subnet mask behind it
- The default router or gateway specifies which server to forward packets to that have an external destination
- The DNS nameserver is the IP address of a server that helps resolving names to IP addresses
- IPv4 is still the most common IP version, but IPv6 addresses can be used as well
- IPv6 addresses are written in hexadecimal notation (fe80::143f:bf00:cd27:a844%n0)
- IPv4 and IPv6 can co-exist on the same network interface

Naming conventions



Interface naming conventions:

- ✓ eno1: Onboard device index from firmware/BIOS
- ✓ ens1: PCIe hotplug slot index
- ✓ enp2s0: Physical location of hardware connector
 - enx525400d5e0fb: MAC address (optional)
- eth0: Traditional kernel naming (fallback) → deprecated

Interface prefixes

- en: ethernet → old prefix was eth
- wl: wireless LAN (WLAN)
- ww: Wireless Wide Area Network (WWAN)
- ib: InfiniBand
- lo: loopback

Core Management Tools



- nmtui (Text User Interface)

- Menu-driven, curses-based interface
- More intuitive for beginners
- No need to remember command syntax

- nmcli (Command Line)

- More powerful and scriptable
- Faster for repeated tasks
- Recommended for advanced users and automation

```
# Install if not present
```

```
sudo dnf install NetworkManager NetworkManager-tui
```

```
# Launch
```

```
nmtui
```

```
# Basic syntax
```

```
nmcli [OPTIONS] OBJECT {COMMAND}
```

```
# Common objects: connection, device, general
```

Basic Operations & Commands



■ Viewing Configuration

```
# Show all connections  
nmcli connection show
```

```
# Show active connections only  
nmcli connection show --active
```

```
# Show detailed device info  
nmcli device show enp0s3
```

```
# Show IP addresses  
ip addr
```

■ Connection Management

```
# Bring connection up/down  
nmcli connection up enp0s3  
nmcli connection down enp0s3
```

```
# Down and up in sequence (minimal downtime)  
nmcli con down enp0s3 && nmcli con up enp0s3
```

```
# Delete a connection  
nmcli connection delete enp0s3
```

```
# Reload all connections  
nmcli connection reload
```



Understanding the ip command

- The **ip** command can be used to manage all aspects of IP networking
- It replaces the legacy ifconfig command [so do not use ifconfig anymore]
- Use **ip addr** to manage address properties
- Use **ip link** to show the link properties
- Use **ip route** to manage the route properties

Setting up IP address



■ Setting up IP address manually

Step 1: Set method to manual (static)

nmcli connection modify "ens160" ipv4.method manual

Step 2: Set IP address, gateway, and DNS

nmcli connection modify "ens160" \
 ipv4.addresses "192.168.50.1/24" \
 ipv4.gateway "192.168.50.254" \
 ipv4.dns "8.8.8.8 1.1.1.1"

Step 3: Apply changes (MUST do this)

nmcli connection down "ens160" && nmcli connection up "ens160"

■ Setting up IP address automatically

To revert to DHCP (automatic IP)

nmcli connection modify "ens160" \
 ipv4.method auto \
 ipv4.addresses "" \
 ipv4.gateway "" \
 ipv4.dns ""

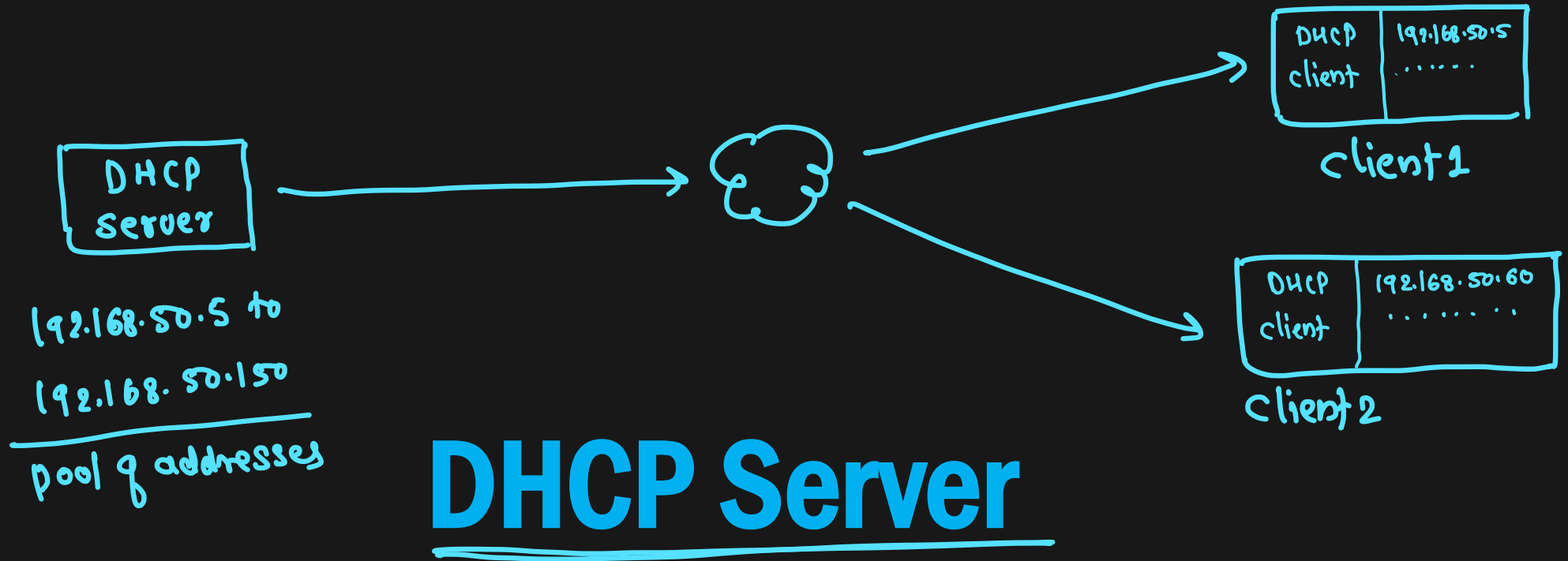
Apply

nmcli connection down "ens160" && nmcli connection up "ens160"

Managing Networks



- **ip link show**: shows the network interfaces
- **ip -s link show**: shows the network interfaces along with statistics
- **ip route**: shows routing table
- **ping <ip>**: used to check the connectivity between hosts
- **traceroute <domain>**: used to see the route taken to reach domain
- **tracpath <domain>**: used to see the path taken to reach domain
- **ss -ta**: used to investigate socket issues
- **nmcli**: used to modify interface settings and save to /etc/sysconfig/network-scripts/
- **dig <domain>**: use to show the dns record of the domain



DHCP



- A network management protocol used to automatically assign IP addresses and other network configuration parameters to devices on a TCP/IP network
- Eliminates the need for manual (static) IP configuration
- Operates at the Application Layer of the OSI model
- Uses UDP as the transport protocol
- Ports
 - DHCP Server: UDP port 67
 - DHCP Client: UDP port 68
- Why DHCP?
 - Automation: Reduces administrative overhead
 - Avoids Conflicts: Prevents two devices from getting the same IP address
 - Efficient IP Use: Dynamically allocates IPs from a pool, releasing unused ones
 - Mobility: Devices get valid IPs automatically when moving between networks
 - Centralized Management: Configuration (DNS, gateway) is managed from a DHCP server

Key Components



- DHCP Server → with DHCP software | process
 - A device (router or server) that holds IP addresses and configuration.
- DHCP Client → software | process
 - Any device requesting an IP (laptop, phone, printer).
- Scope → pool of addresses
 - A range of IP addresses available for lease to clients.
- Lease
 - The time duration for which an IP is assigned to a client.
- Reservation → MAC binding
 - A permanent lease (static mapping) of an IP to a specific MAC address.
- Options
 - Additional parameters (e.g., default gateway, DNS server, domain name).

DHCP Messages



■ DHCPDISCOVER

- Client to Server (broadcast)
- Client searches for a DHCP server

■ DHCPOFFER

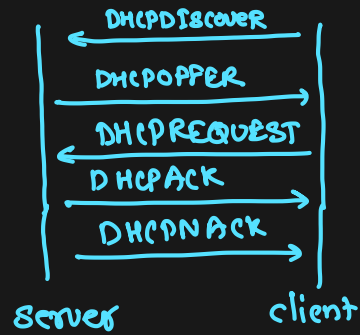
- Server to Client (unicast/broadcast)
- Server offers an available IP address

■ DHCPREQUEST

- Client to Server (broadcast)
- Client accepts the offer

■ DHCPACK

- Server to Client (unicast)
- Server confirms assignment and provides lease + options



■ DHCPNAK

- Server to Client
- Denies the request (e.g., IP no longer available or invalid)

■ DHCPDECLINE

- Client to Server
- Client detects IP is already in use (e.g., via ARP)

■ DHCPRELEASE

- Client to Server
- Client voluntarily gives up its IP address

■ DHCPINFORM

- Client to Server
- Client already has an IP (manually) but needs other options (DNS, gateway)

DHCP Lease Lifecycle



- Initialization (Bootting)
 - Client sends DISCOVER: obtains IP (DORA)
- Normal Operation
 - Client uses the IP → lease time
- Renewal (T1 - 50% of lease time)
 - Client sends unicast REQUEST to original server to renew lease
- Rebinding (T2 - 87.5% of lease time)
 - If renewal fails, client broadcasts REQUEST (any server can respond)
- Expiration
 - If no response, client stops using IP and starts DORA over

Best Practices



- Only one DHCP server per network - Disable DHCP on your router to avoid conflicts
- Server needs static IP - DHCP servers should not obtain their own IP via DHCP
- Plan your IP range - Ensure your DHCP pool doesn't overlap with static IPs used by servers, printers, or network equipment
- Backup configuration - Always backup /etc/dhcp/dhcpd.conf before making changes
- Monitor lease exhaustion - Use tools like dhcpd-pools to monitor IP usage

dhcp server



■ Configure server IP address

- `sudo nmcli connection modify "ens160" ipv4.address <ip>`
- `sudo nmcli connection modify "ens160" ipv4.method manual`
- `sudo nmcli connection down "ens160"`
- `sudo nmcli connection up "ens160"`
- `ip address show`

■ Install dhcpd server

- `sudo dnf update -y`
- `sudo yum install -y dhcp-server`

■ Configure server (/etc/dhcp/dhcpd.conf)

```
default-lease-time 600;  
max-lease-time 7200;  
authoritative;  
subnet 192.168.50.0 netmask 255.255.255.0 {  
    range 192.168.50.100 192.168.50.200;  
    option routers 192.168.50.1;  
    option subnet-mask 255.255.255.0;  
    option domain-name-servers 8.8.8.8;  
}
```

■ Validate configuration

- `sudo dhcpd -t`

■ Bind dhcp with interface

- `sudo vim /etc/sysconfig/dhcpd`
- `DHCPDARGS=ens160`

■ Start server

- `sudo systemctl enable --now dhcpd`

■ Firewall settings

- `sudo firewall-cmd --add-service dhcp --permanent`
- `sudo firewall-cmd --reload`